

CS501

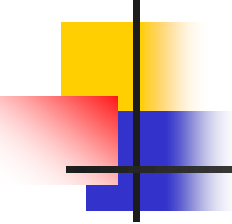
Introductory Computation for Scientists



Dr. Ash Pahwa

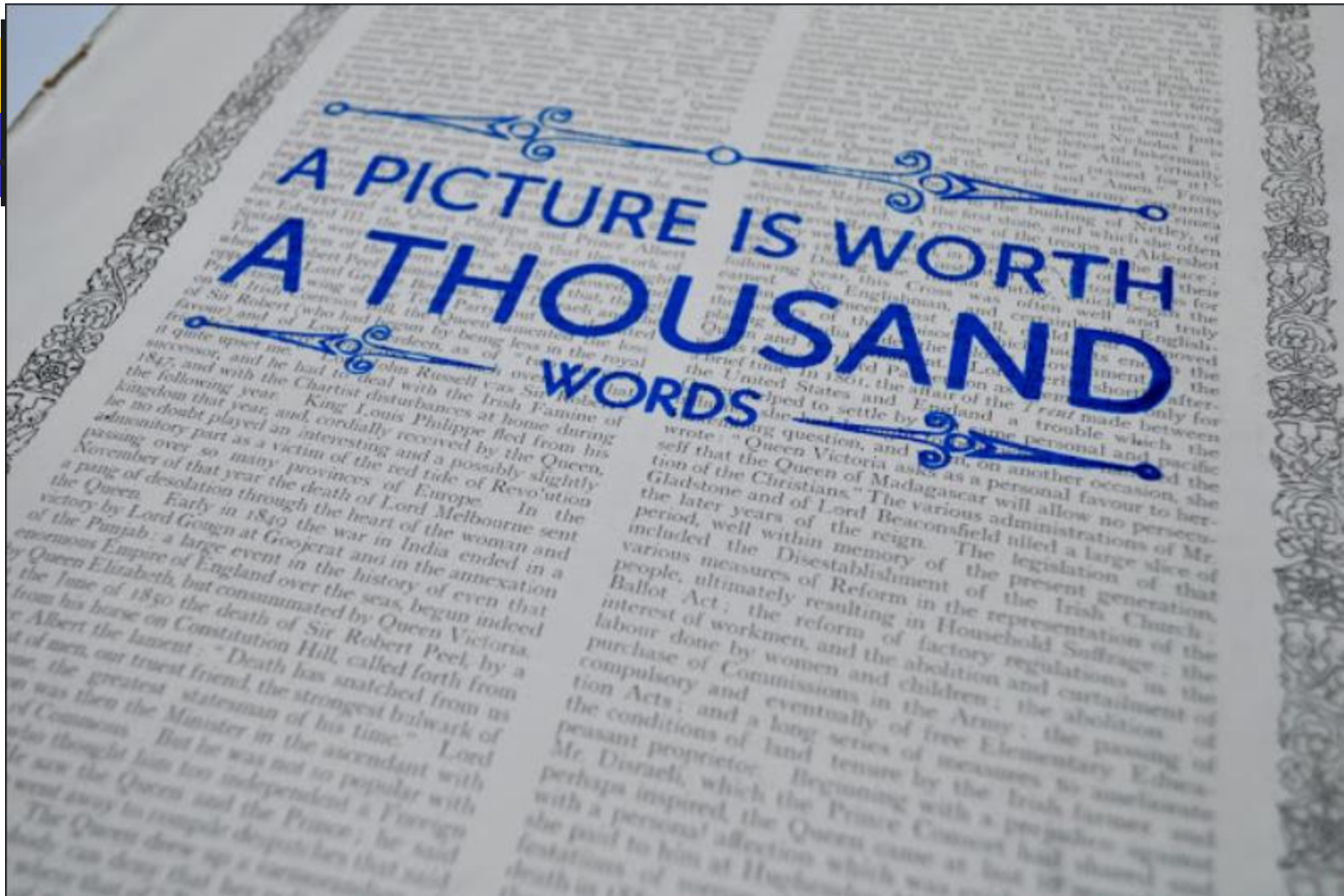
Lesson 8: Data Visualization

Lesson 8.1: Importance of Data Visualization in Machine Learning



Outline

- Importance of Data Visualization
- Data Visualization Tools
 - R: Basic Graphics
 - R: ggplot2
 - Python: Matplotlib
 - Python: Seaborn
 - Tableau + Power BI



Charles Joseph Minard

1781 – 1870

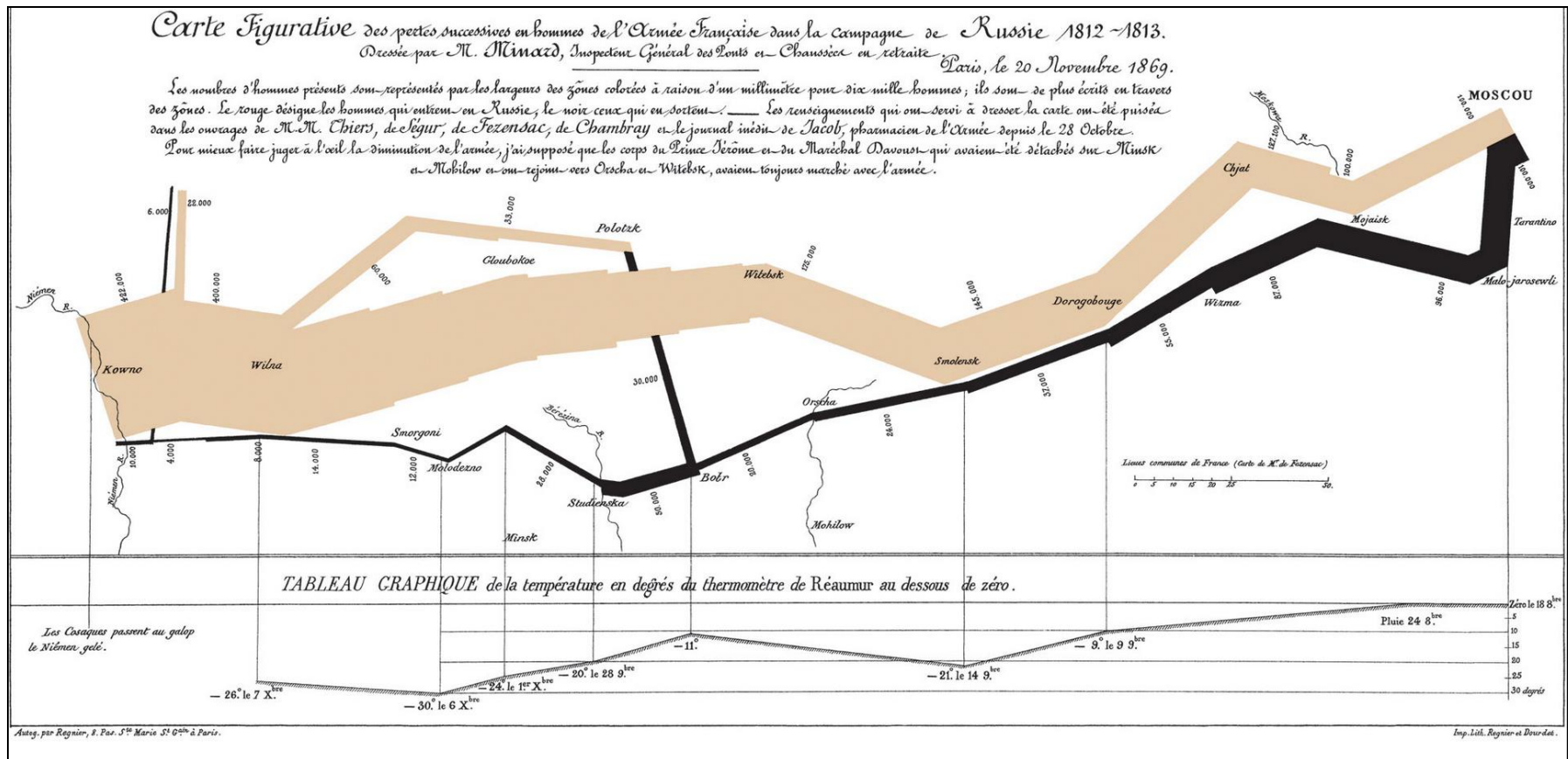


- French civil engineer who created one of the greatest graphs of all time
- Visualized Napoleon's 1812 invasion of Russia, including
 - the number of soldiers
 - the direction of the march
 - the latitude and longitude of each city
 - the temperature on the return journey
 - Dates in November and December

Most Important Step

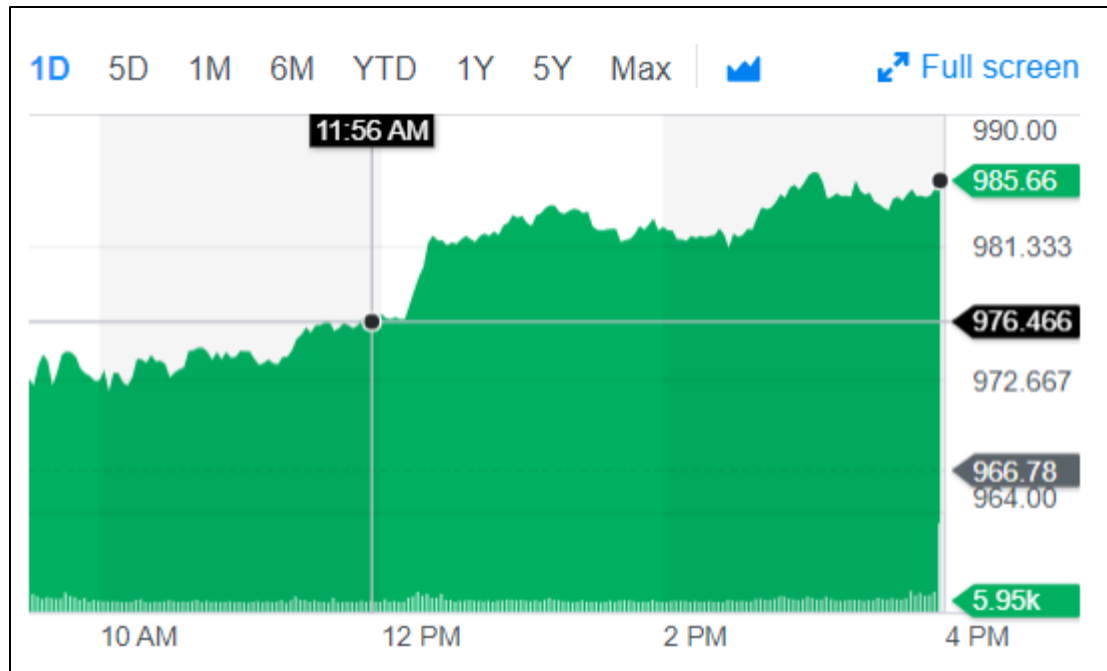
Communicate Your Results: Tell a Story

Napoleon March To Moscow



Interactive Graphics

- <https://finance.yahoo.com/quote/GOOGL?p=GOOGL>



Spurious Correlations

- <http://www.tylervigen.com/spurious-correlations>

tylervigen.com [about](#) | [twitter](#) | [email](#) | [subscribe](#)

Spurious correlations



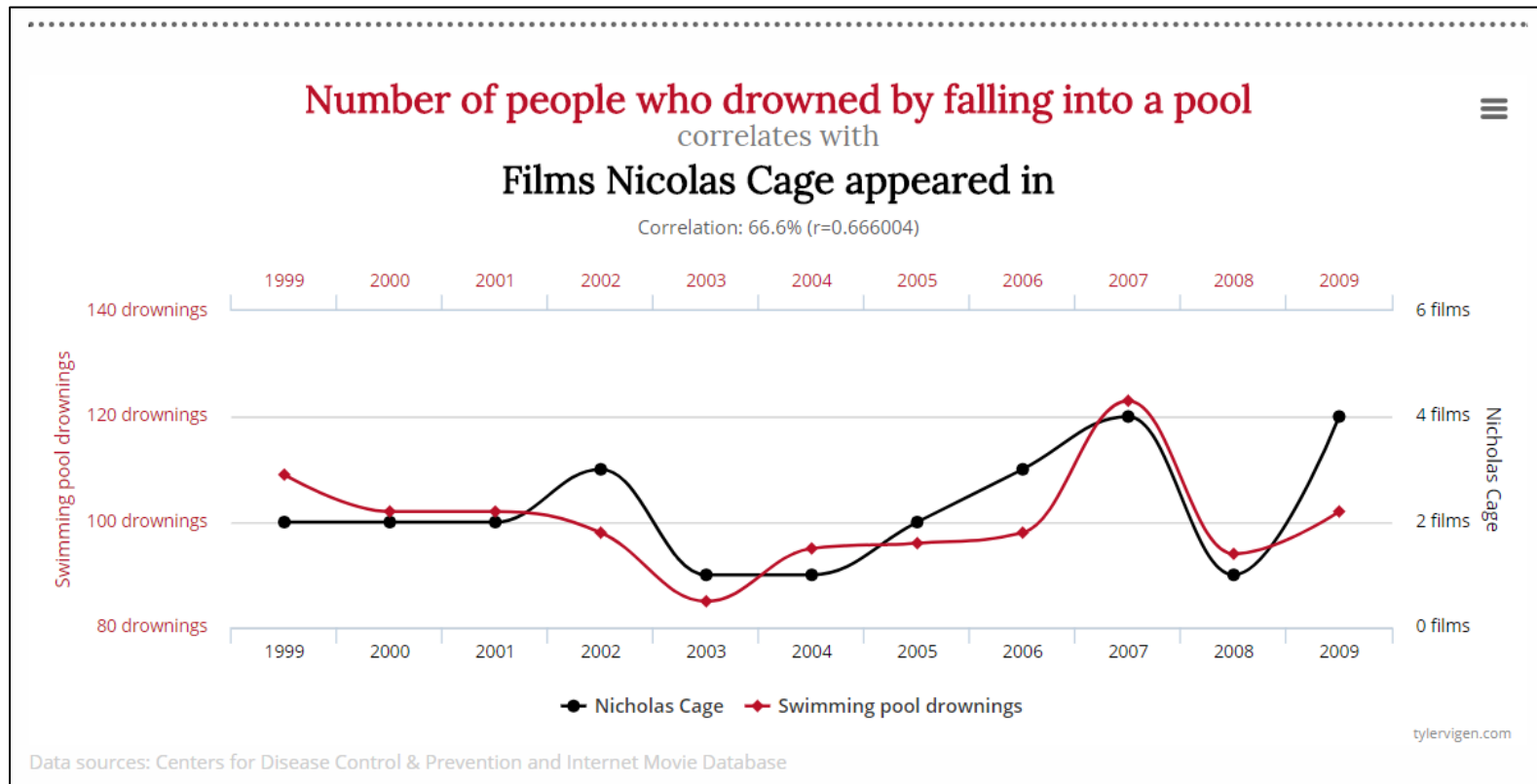
Now a ridiculous book!

- Spurious charts
- Fascinating factoids
- Commentary in the footnotes

[Amazon](#) | [Barnes & Noble](#) | [Indie Bound](#)

Spurious Correlations

Interactive Graphics



Prof. David Donoho

50 Years of Data Science

Teaching Data Science



David Donoho
[Statistics Department](#)
[Stanford University](#)
Stanford, CA 94305, USA
donoho@stat.stanford.edu
Voice: (650) 723-3350
Fax: (650) 725-8977

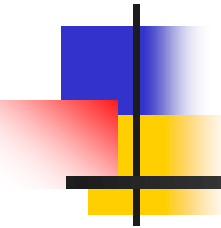
- UC Berkeley
- Master's in Data Science
 - Exploring and Analyzing Data
 - Storing and Retrieving data
 - Applied Machine Learning
 - **Data Visualization and Communication**
 - Research Design and Application for Data and Analysis

- Donoho's Proposal
 - Data Exploration and Preparation
 - Data Representation and Transformation
 - Computing with Data
 - Data Modeling
 - **Data Visualization and Presentation**
 - Science about Data Science

Visitors only Read 20% of a Web Page

- Heat Map of a web page can be created using the mouse movement



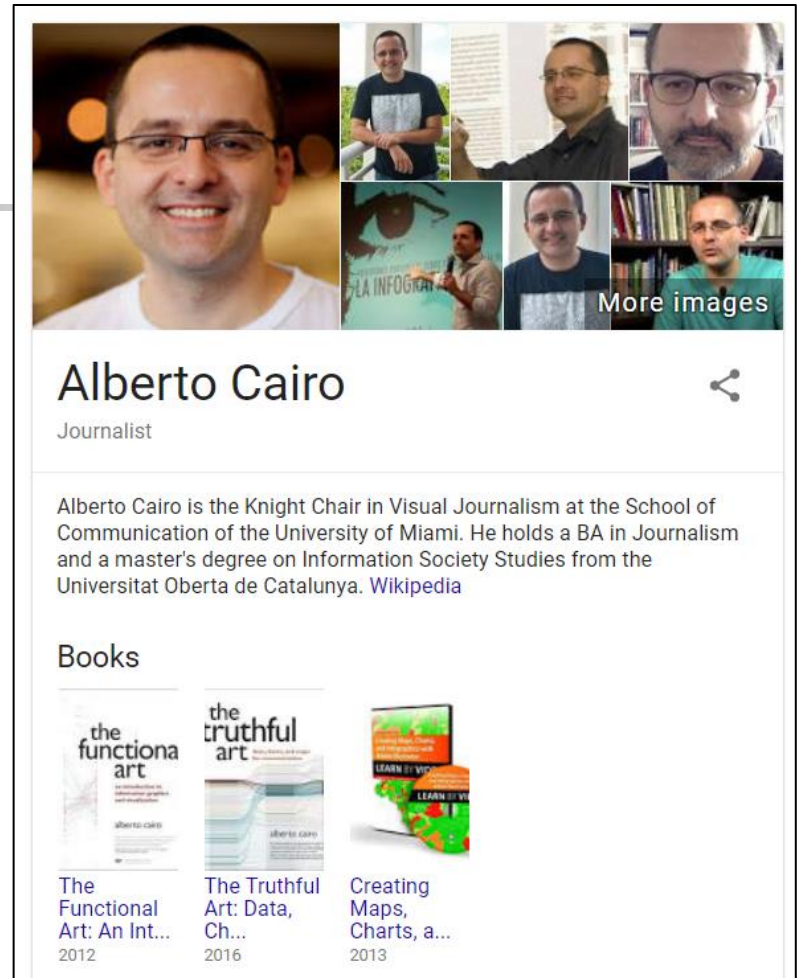


Data Visualization in Python

Matplotlib
Seaborn

Alberto Cairo

- Alberto Cairo is the Knight Chair in Visual Journalism at the School of Communication of the University of Miami.
- Cairo worked as a data-journalist and information designer at El Mundo, Spain's second largest printed daily newspaper, and he was the director for Infographics and Multimedia at Editora Globo in Brazil.
- He is author of
 - "The Functional Art: an Introduction to Information Graphics and Visualization" (2012) and
 - "The Truthful Art: Data, Charts, and Maps for Communication" (2016).



Alberto Cairo

Journalist

Alberto Cairo is the Knight Chair in Visual Journalism at the School of Communication of the University of Miami. He holds a BA in Journalism and a master's degree on Information Society Studies from the Universitat Oberta de Catalunya. [Wikipedia](#)

Books



The Functional Art: An Int...
2012



The Truthful Art: Data, Ch...
2016



Creating Maps, Charts, a...
2013

Edward Tufte

- Edward Rolf Tufte is an American statistician and professor emeritus of political science, statistics, and computer science at Yale University.
- He is noted for his writings on information design and as a pioneer in the field of data visualization.

Edward Tufte



Tufte during his one-day course in Dallas, May 21, 2015

Born	March 14, 1942 (age 75) ^[citation needed] Kansas City, Missouri
Occupation	Professor, statistician, writer, sculptor
Nationality	American
Notable works	• <i>The Visual Display of Quantitative Information</i> • <i>Beautiful Evidence</i>

John Hunter

- John D. Hunter (1968 – 2012) was an American neurobiologist and the original author of matplotlib.

John D. Hunter

Author



John D. Hunter was an American neurobiologist and the original author of matplotlib. [Wikipedia](#)

Born: August 1, 1968, [Dyersburg, TN](#)

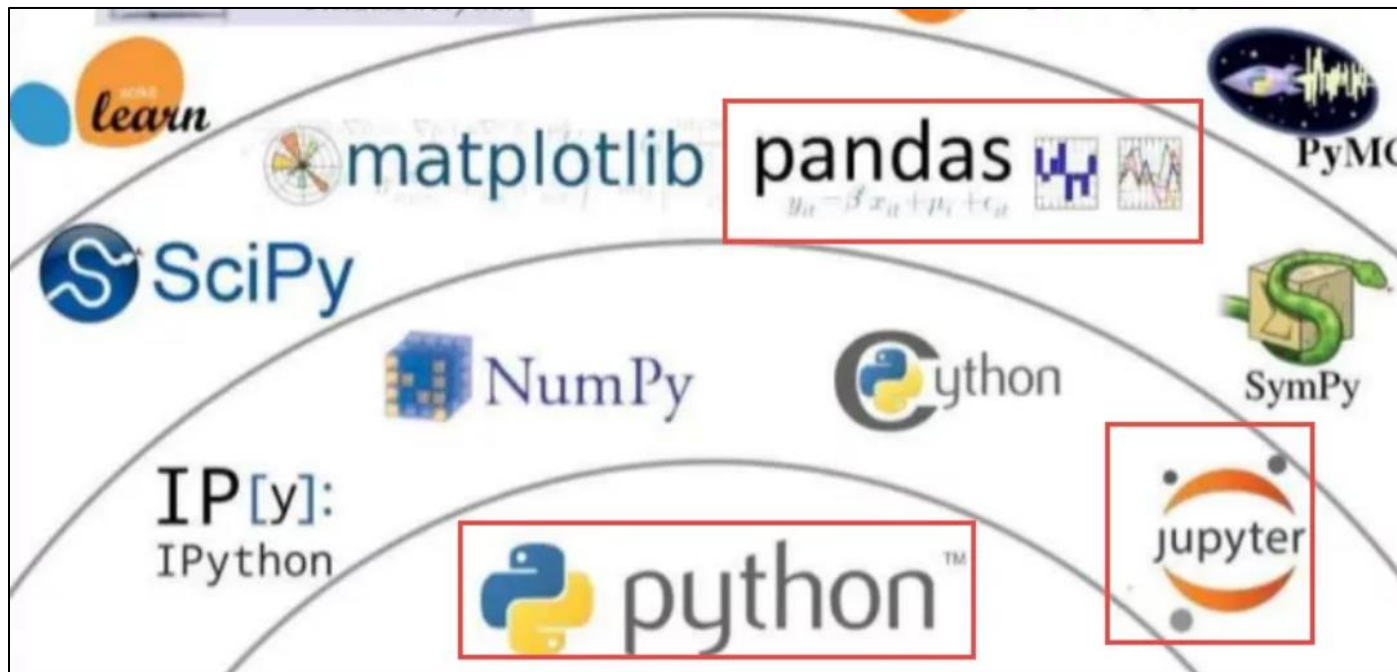
Died: August 28, 2012, [Chicago, IL](#)

Alma mater: [University of Chicago](#)

Known for: [matplotlib](#)

Books: [Interview Success](#)

What is Matplotlib?



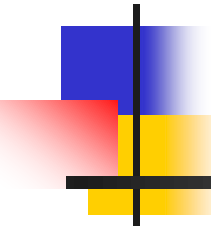


Tableau + Power BI

Pat Hanrahan & Chris Stolte



Pat Hanrahan

Computer researcher

Pat Hanrahan is a computer graphics researcher, the Canon USA Professor of Computer Science and Electrical Engineering in the Computer Graphics Laboratory at Stanford University. [Wikipedia](#)

Born: 1954

Education: [University of Wisconsin-Madison](#) (1985)

Organizations founded: [Tableau Software](#)

Notable students: [Maneesh Agrawala](#), [Tamara Munzner](#)

Awards: [Academy Award for Best Technical Achievement](#), [Academy Scientific and Engineering Award](#)

Academic advisor: [Antony Stretton](#)



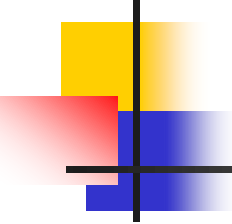
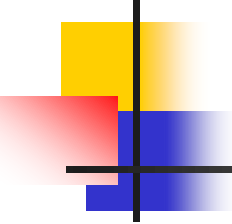


Tableau Company

- The company was founded in Mountain View, California in January, 2003 by Chris Stolte, Christian Chabot and Pat Hanrahan. Their initial aim was to commercialize research conducted at Stanford University's Department of Computer Science between 1999 and 2002.
- Professor Pat Hanrahan and Ph.D. student Chris Stolte who specialized in visualization techniques for exploring and analyzing relational databases and data cubes[[] led research in the use of table-based displays to browse multidimensional relational databases.



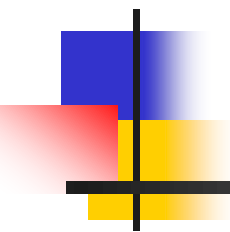
Power BI

New Product Released in 2015

- Power BI
 - Similar to Tableau
- Data can be extracted from any place on the web
- Visuals are based on HTML5 and can be seen by any browser or any mobile device

Statistical Computing and Data Visualization

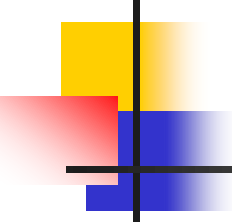
DSO 545



Dr. Ash Pahwa

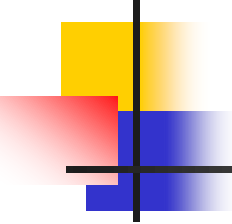
Lesson 8: Bar + Pie + Scatter Plots

Lesson 8.2: Bar Plots



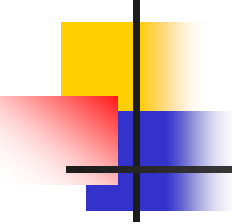
Plot Characteristics

- Change the Figure Size
- Adding Colors + Color Map
- Adding labels on plots
- Add Legend
- Font Size
- Set Axis Range
- Change Tick Frequency
- Rotate Axis Tick Labels
- Draw Vertical and Horizontal Lines on Plot



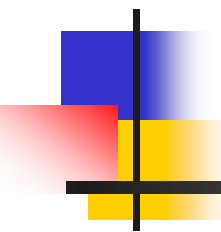
Type of Plots

1. Line Plot
2. Bar Plot
3. Pie Chart
4. Scatter Plot
5. Histogram Plot
6. Box Plot
7. Joint Plot
8. Violin Plots
9. Stack Plots
10. Heat Maps
11. Spectrogram Plots
12. 3D Plots
13. Contour Plots
14. KDE Plots
15. Interactive Plots



Outline

- Bar Plot: Categorical Variable + Numerical Variable
- Adding Labels in a Plot
- Adding Straight Lines on the Plot to Display Threshold
- Adding Color to the Plot
- Horizontal Bar Plots



Bar Plot

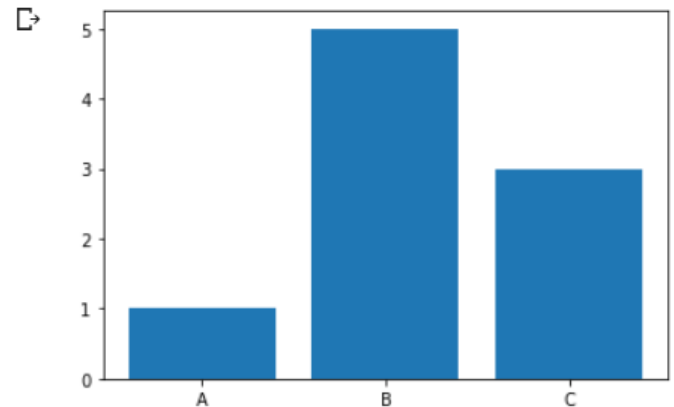
Bar Plot

If one of the variables is Categorical,
bar plot is created automatically

```
[1] #1  
import numpy as np  
import matplotlib.pyplot as plt
```

```
[2] #page 189  
#  
x = ['A', 'B', 'C']  
y = [1, 5, 3]
```

```
plt.bar(x, y)  
plt.show()
```



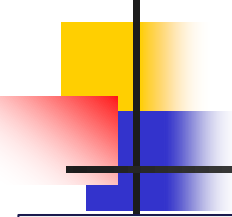
Data File: World Happiness Report - 2021

	A	B	C	D	E	F	G	H	I	J
1	Country name	Regional indicator	Ladder score	Standard error	upperwhis	lowerwhis	Logged GDP	Social support	Healthy life expectancy	Freedom to make life choices
2	Finland	Western Europe	7.842	0.032	7.904	7.78	10.775	0.954	72	0.949
3	Denmark	Western Europe	7.62	0.035	7.687	7.552	10.933	0.954	72.7	0.946
4	Switzerland	Western Europe	7.571	0.036	7.643	7.5	11.117	0.942	74.4	0.919
5	Iceland	Western Europe	7.554	0.059	7.67	7.438	10.878	0.983	73	0.955
6	Netherlands	Western Europe	7.464	0.027	7.518	7.41	10.932	0.942	72.4	0.913
7	Norway	Western Europe	7.392	0.035	7.462	7.323	11.053	0.954	73.3	0.96
8	Sweden	Western Europe	7.363	0.036	7.433	7.293	10.867	0.934	72.7	0.945
9	Luxembourg	Western Europe	7.324	0.037	7.396	7.252	11.647	0.908	72.6	0.907
10	New Zealand	North America and ANZ	7.277	0.04	7.355	7.198	10.643	0.948	73.4	0.929
11	Austria	Western Europe	7.268	0.036	7.337	7.198	10.906	0.934	73.3	0.908
12	Australia	North America and ANZ	7.183	0.041	7.265	7.102	10.796	0.94	73.9	0.914
13	Israel	Middle East and North Africa	7.157	0.034	7.224	7.09	10.575	0.939	73.503	0.8
14	Germany	Western Europe	7.155	0.04	7.232	7.077	10.873	0.903	72.5	0.875
15	Canada	North America and ANZ	7.103	0.042	7.185	7.021	10.776	0.926	73.8	0.915

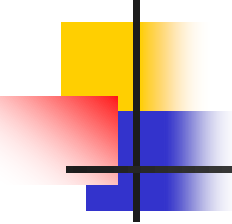
(Categorical)

Freedom to Make life Choices

(Numerical)



	A	B	C	D	E	F	G	H	I	J	K
1	Country name	Regional indicator	Ladder sco	Standard e	upperwhis	lowerwhis	Logged GD	Social supp	Healthy lif	Freedom to make life choices	Ge
2	Finland	Western Europe	7.842	0.032	7.904	7.78	10.775	0.954	72	0.949	
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11	Austria	Western Europe	7.268	0.036	7.337	7.198	10.906	0.934	73.3	0.908	
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15	Canada	North America and ANZ	7.103	0.042	7.185	7.021	10.776	0.926	73.8	0.915	
16	Ireland	Western Europe	7.085	0.04	7.164	7.006	11.342	0.947	72.4	0.879	

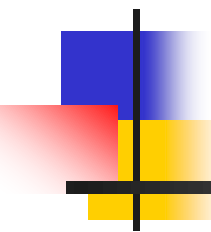


Read Data File

```
[4] from google.colab import files
    f = files.upload()

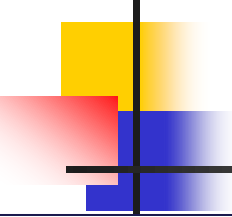
Choose Files world-happi...rt-2021.csv
• world-happiness-report-2021.csv(text/csv) - 21687 bytes, last modified: 3/23/2023 - 100% done
Saving world-happiness-report-2021.csv to world-happiness-report-2021.csv

import pandas as pd
df = pd.read_csv('world-happiness-report-2021.csv')
```



Adding Labels of Categorical Variable

(Categorical) Freedom to Make life Choices (Numerical)

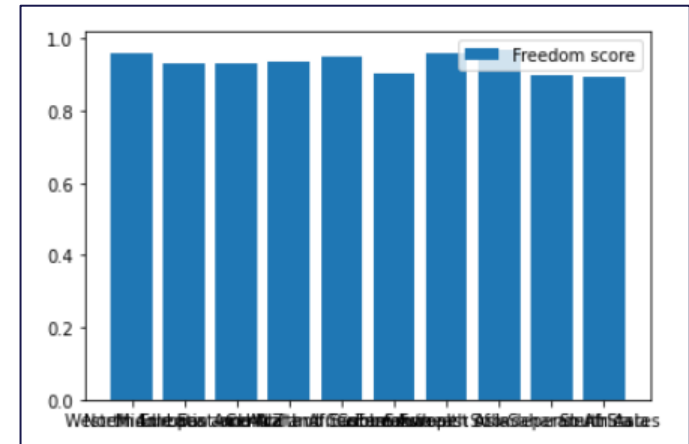


#Page 191

```
fig, ax = plt.subplots()

ax.bar(df['Regional indicator'], df['Freedom to make life choices'], label='Freedom score')
ax.legend()
plt.show()
```

X labels are written over



(Categorical) Freedom to Make life Choices (Numerical)

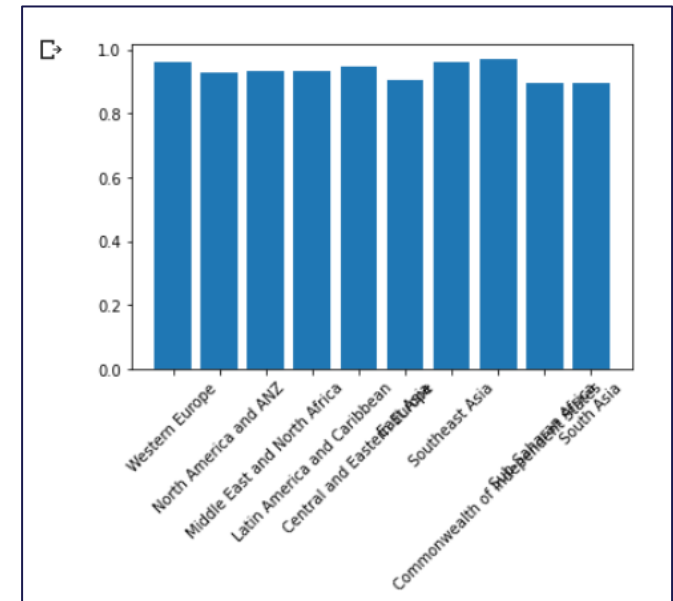
```
#Page 191

fig, ax = plt.subplots()

ax.bar(df['Regional indicator'], df['Freedom to make life choices'], label='Freedom score')
#ax.legend()

plt.xticks(rotation=45, wrap=False)
plt.show()
```

Rotate the x labels by 45 degrees – doesn't work



(Categorical) Freedom to Make life Choices (Numerical)

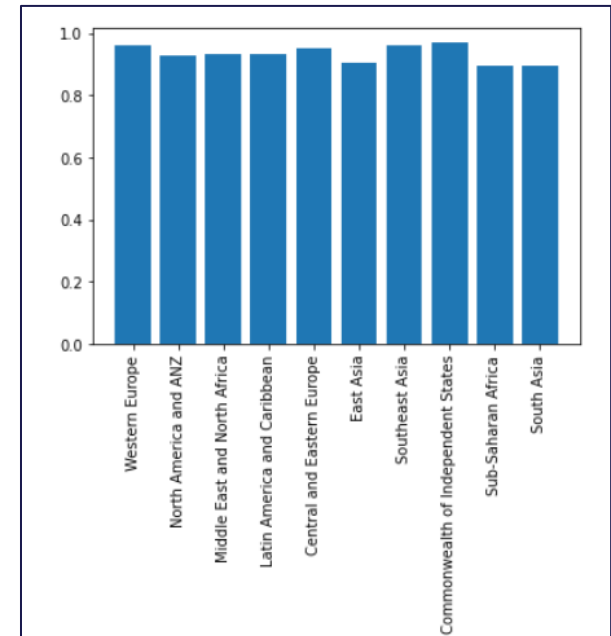
#Page 191

```
fig, ax = plt.subplots()
```

```
ax.bar(df['Regional indicator'], df['Freedom to make life choices'], label='Freedom score')  
#ax.legend()
```

```
plt.xticks(rotation=90, wrap=False)  
plt.show()
```

Rotate the x labels by 90 degrees





Adding Straight Lines on the Plot to Display Threshold

Draw Mean and Median Line on Plot



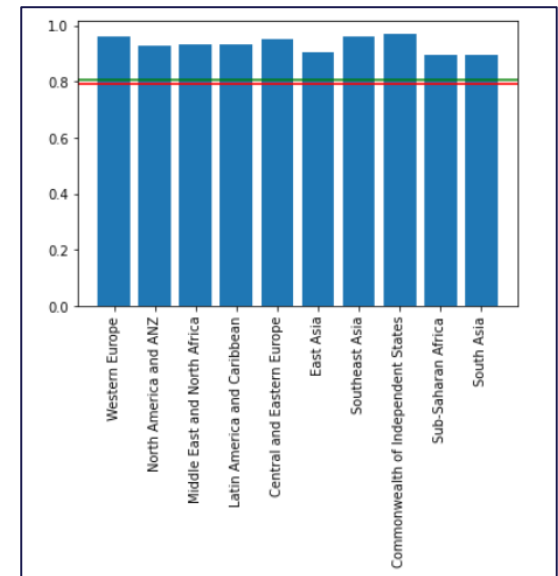
Page 192

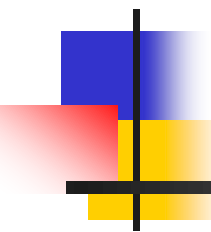
```
mean = df['Freedom to make life choices'].mean()
median = df['Freedom to make life choices'].median()

fig, ax = plt.subplots()
ax.bar(df['Regional indicator'], df['Freedom to make life choices'], label='Freedom score')
plt.xticks(rotation=90, wrap=False)

ax.axhline(mean,0,1, color='red', label='Mean Value for Dataset')
ax.axhline(median,0,1, color='green', label='Median Value for Dataset')

#ax.legend()
plt.show()
```





Adding Color to the Plot

Groupby: Regional indicator: Compute mean of each group

Create Bar Plot of each group with colors

```
[14] #page 198
df_means = df.groupby('Regional indicator').mean()
#print(df_means)

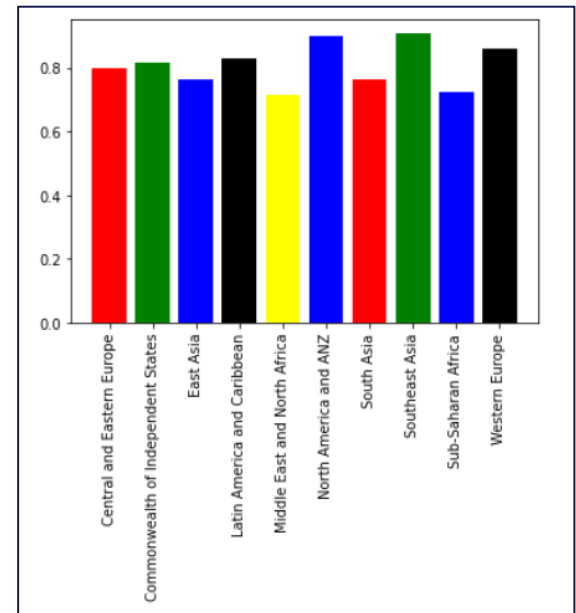
fig, ax = plt.subplots()

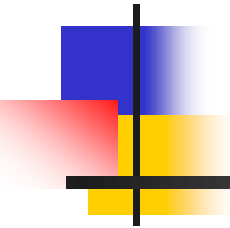
# colors = np.arange(len(df['Regional indicator']))
# print(colors)

colors = ['red','green','blue','black','yellow','blue']

ax.bar(df_means.index, df_means['Freedom to make life choices'], label = 'Freedom score', color = colors)

plt.xticks(rotation=90, wrap=False)
plt.show()
```





Horizontal Bar Plot

Horizontal Bars with Color

#page 199

```
fig, ax = plt.subplots()

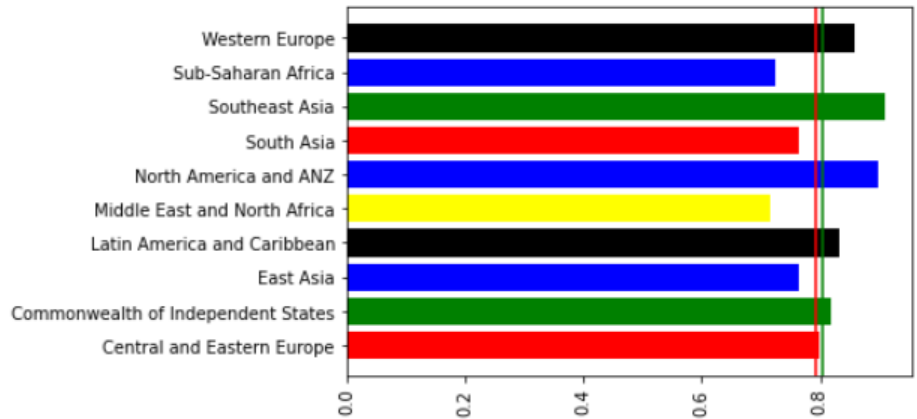
colors = ['red', 'green', 'blue', 'black', 'yellow', 'blue']

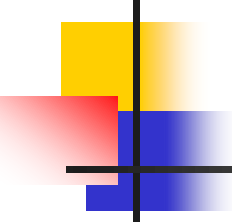
ax.barh(df_means.index, df_means['Freedom to make life choices'], label = 'Freedom score', color = colors)

plt.xticks(rotation=90, wrap=False)

ax.axvline(mean,0,1, color='red', label='Mean Value for Dataset')
ax.axvline(median,0,1, color='green', label='Median Value for Dataset')

#ax.legend()
plt.show()
```



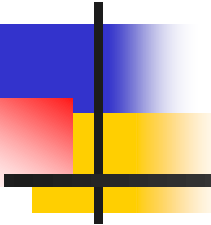


Summary

- Bar Plot: Categorical Variable + Numerical Variable
- Adding Labels in a Plot
- Adding Straight Lines on the Plot to Display Threshold
- Adding Color to the Plot
- Horizontal Bar Plots

Statistical Computing and Data Visualization

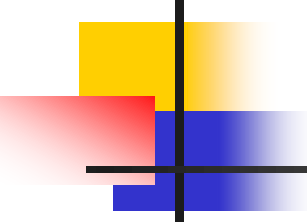
DSO 545



Dr. Ash Pahwa

Lesson 21: Bar + Pie + Scatter Plots

Lesson 21.2: Pie Plots



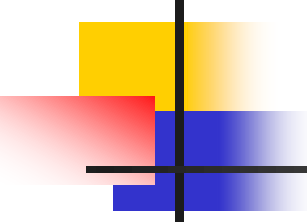
Plot Characteristics

- Change the Figure Size
- Adding Colors + Color Map
- Adding labels on plots
- Add Legend
- Font Size
- Set Axis Range
- Change Tick Frequency
- Rotate Axis Tick Labels
- Draw Vertical and Horizontal Lines on Plot



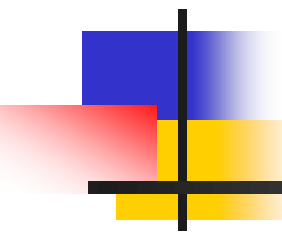
Type of Plots

1. Line Plot
2. Bar Plot
3. Pie Chart
4. Scatter Plot
5. Histogram Plot
6. Box Plot
7. Joint Plot
8. Violin Plots
9. Stack Plots
10. Heat Maps
11. Spectrogram Plots
12. 3D Plots
13. Contour Plots
14. KDE Plots
15. Interactive Plots



Outline

- Pie Plots
- Add Labels to a Pie Plot
- Using Specific Colors to a Pie Plot
- Adding Percentage to a Pie Plot
- Exploding Wedges of a Pie Plot



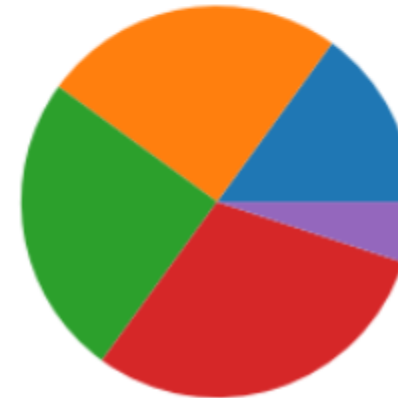
Pie Plots

A Simple Pie Plot

```
[1] #1
import numpy as np
import matplotlib.pyplot as plt
```

```
x = [15, 25, 25, 30, 5]

fig, ax = plt.subplots()
ax.pie(x)
plt.show()
```



Add Labels to a Pie Plot



```
x = [15, 25, 25, 30, 5]
```

```
labels = ['Very Likely', 'Likely', 'Unsure', 'Unlikely', 'Very Unlikely']
```

```
fig, ax = plt.subplots()
```

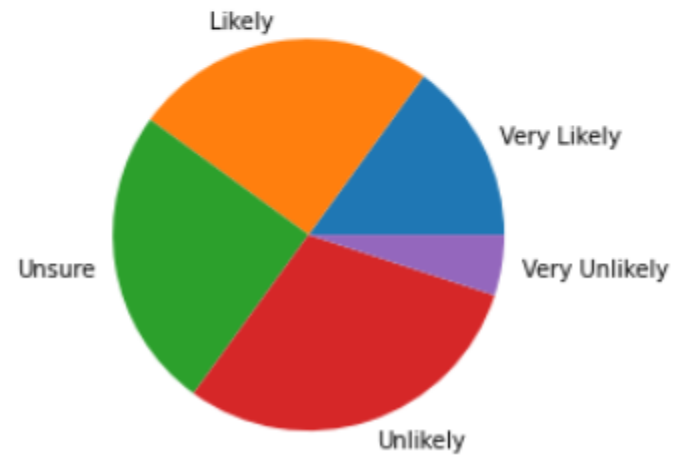
```
ax.pie(x, labels = labels)
```

```
ax.set_title('Survey responses')
```

```
plt.show()
```



Survey responses

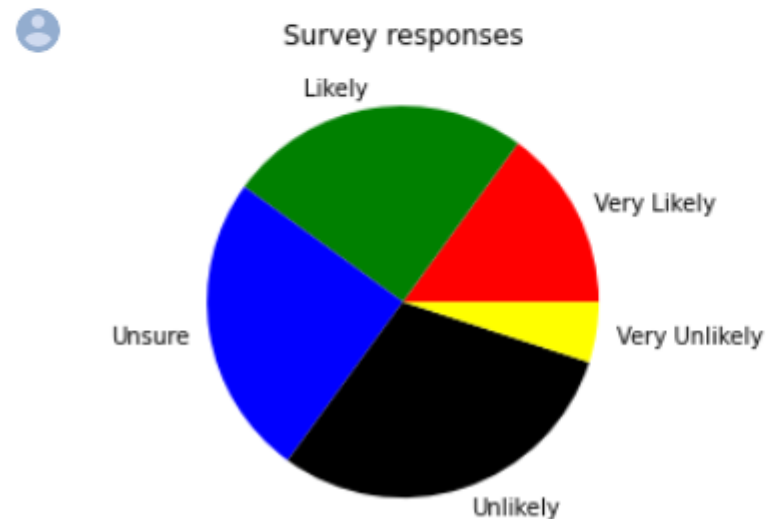


Using Specific Colors to a Pie Chart

```
x = [15, 25, 25, 30, 5]

labels = ['Very Likely', 'Likely', 'Unsure', 'Unlikely', 'Very Unlikely']
colors = ['red', 'green', 'blue', 'black', 'yellow', 'blue']

fig, ax = plt.subplots()
ax.pie(x, labels = labels, colors = colors)
ax.set_title('Survey responses')
plt.show()
```

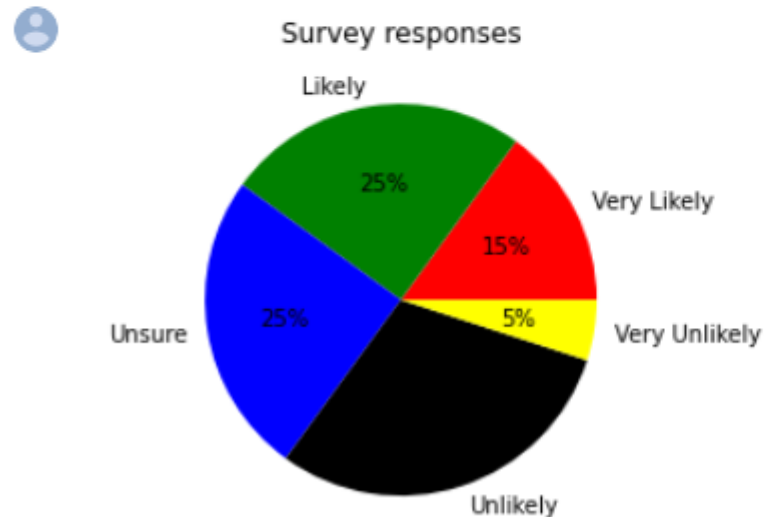


Adding Percentage on the Plot Itself

```
x = [15, 25, 25, 30, 5]

labels = ['Very Likely', 'Likely', 'Unsure', 'Unlikely', 'Very Unlikely']
colors = ['red', 'green', 'blue', 'black', 'yellow', 'blue']

fig, ax = plt.subplots()
ax.pie(x, labels = labels, colors = colors, autopct='%.0f%%')
ax.set_title('Survey responses')
plt.show()
```



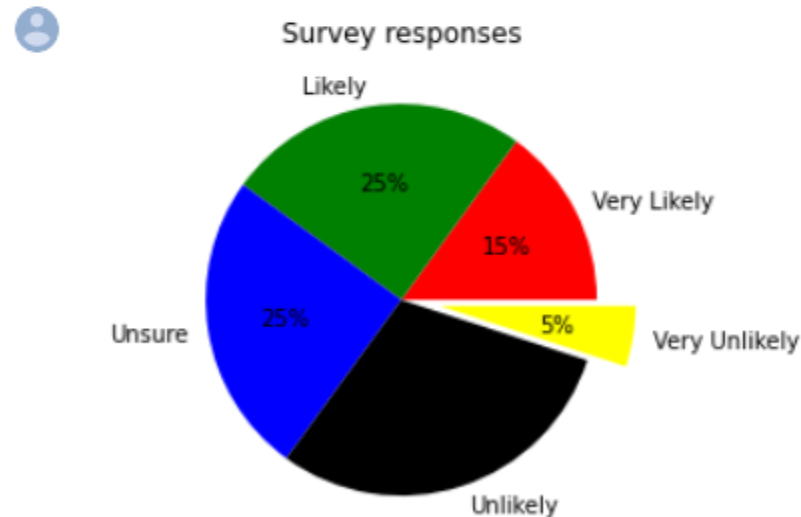
Explode Wedges of a Pie Slice

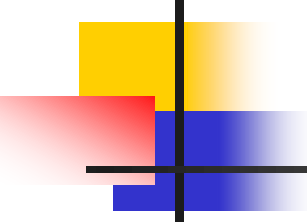
```
x = [15, 25, 25, 30, 5]

labels = ['Very Likely', 'Likely', 'Unsure', 'Unlikely', 'Very Unlikely']
colors = ['red', 'green', 'blue', 'black', 'yellow', 'blue']
explode = [0, 0, 0, 0, 0.2]

fig, ax = plt.subplots()
ax.pie(x, labels = labels, colors = colors, autopct='%0f%%', explode = explode)
ax.set_title('Survey responses')

plt.show()
```



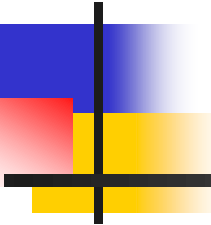


Summary

- Pie Plots
- Add Labels to a Pie Plot
- Using Specific Colors to a Pie Plot
- Adding Percentage to a Pie Plot
- Exploding Wedges of a Pie Plot

Statistical Computing and Data Visualization

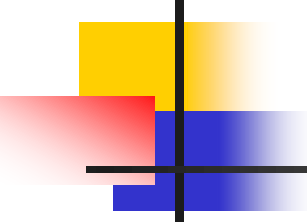
DSO 545



Dr. Ash Pahwa

Lesson 21: Bar + Pie + Scatter Plots

Lesson 21.3: Scatter Plots



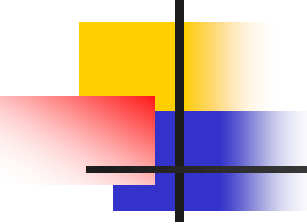
Plot Characteristics

- Change the Figure Size
- Adding Colors + Color Map
- Adding labels on plots
- Add Legend
- Font Size
- Set Axis Range
- Change Tick Frequency
- Rotate Axis Tick Labels
- Draw Vertical and Horizontal Lines on Plot



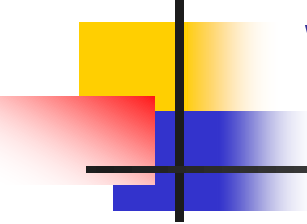
Type of Plots

1. Line Plot
2. Bar Plot
3. Pie Chart
4. Scatter Plot
5. Histogram Plot
6. Box Plot
7. Joint Plot
8. Violin Plots
9. Stack Plots
10. Heat Maps
11. Spectrogram Plots
12. 3D Plots
13. Contour Plots
14. KDE Plots
15. Interactive Plots



Outline

- When do we need to use Scatter Plots?
- Data Set: Ames Iowa Housing
- Create Scatter Plot: Living Area + Price of House
- Create 3D plot Using Scatter/Bubble Plot + Color
 - GDP & Life Expectance of Countries



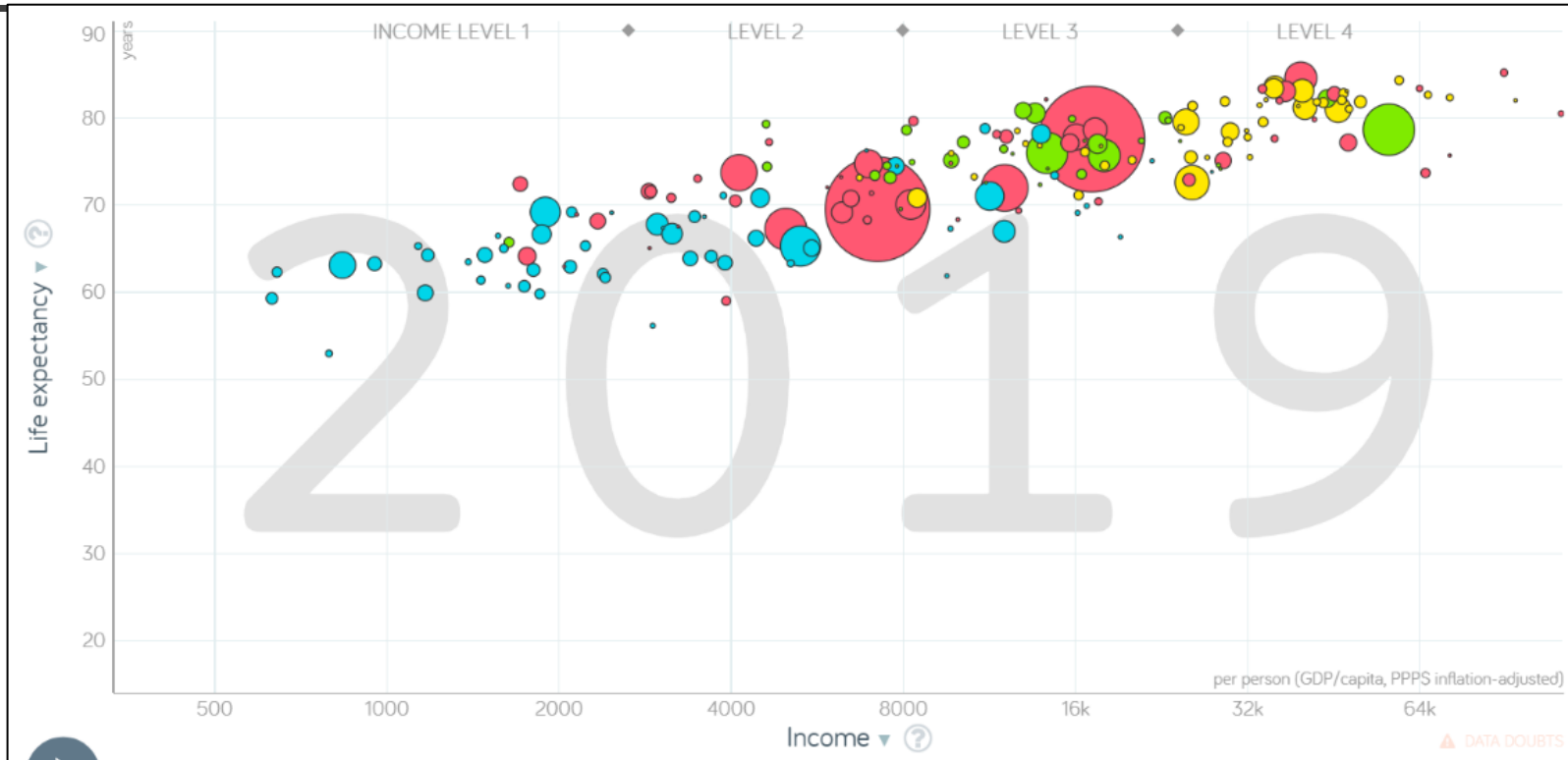
When do we need to Use Scatter Plots?

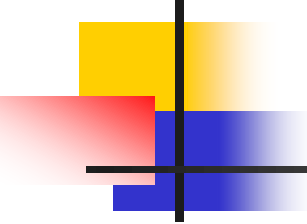
- Scatter plots visualize the relationship between 2 (or 3) numerical features of a single entity
- Examples:
 - House (Entity)
 - Square foot area
 - Price
 - Student (Entity)
 - GRE
 - GPA
 - Country (Entity)
 - Per-capita income
 - Life expectancy

Example: Scatter Plot

Income & Life Expectancy of all countries

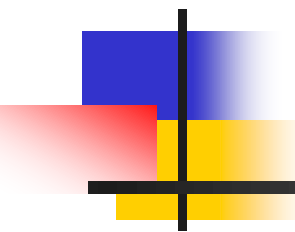
[https://www.gapminder.org/tools/#\\$chart-type=bubbles](https://www.gapminder.org/tools/#$chart-type=bubbles)





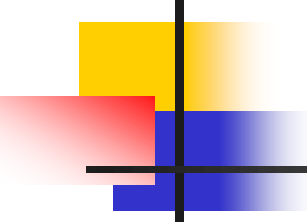
Number of Dimensions that Can be Displayed by Scatter Plot

- Scatter plots visualize the relationship between 2 (or 3) numerical features of a single entity
- X-axis represents 1st dimension
- Y-axis represents 2nd dimension
- The 3rd dimension can be the size of the bubble

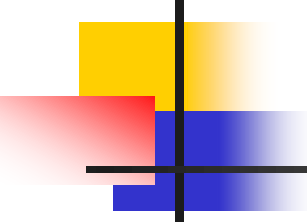


Data Set

Dataset: Ames (City in Iowa) Housing Data



- The Ames Housing dataset, basis of an ongoing Kaggle competition and assigned to bootcamp students globally, is a modern classic.
- It presents data on 81 features of houses -- mostly single-family suburban dwellings -- that were sold in Ames, Iowa in the period 2006-2010, which encompasses the housing crisis.



Load Libraries: Upload Data File

```
[1] #1
import numpy as np
import matplotlib.pyplot as plt
```

```
[2] from google.colab import files
f = files.upload()
```

AmesHousing.csv

- **AmesHousing.csv**(text/csv) - 963738 bytes, last modified: 3/24/2023 - 100% done
Saving AmesHousing.csv to AmesHousing.csv

```
[3] import pandas as pd
df = pd.read_csv('AmesHousing.csv')
```

Read Data File

```
[3] import pandas as pd
     df = pd.read_csv('AmesHousing.csv')
```

```
print(df.head())
```

```
Order  PID  MS SubClass  MS Zoning  Lot Frontage  Lot Area  Street \
0      1  526301100      20      RL      141.0      31770  Pave
1      2  526350040      20      RH       80.0      11622  Pave
2      3  526351010      20      RL       81.0      14267  Pave
3      4  526353030      20      RL       93.0      11160  Pave
4      5  527105010      60      RL       74.0      13830  Pave

Alley  Lot Shape  Land Contour  ... Pool Area  Pool QC  Fence Misc Feature \
0  NaN      IR1      Lvl ...      0      NaN      NaN      NaN
1  NaN      Reg      Lvl ...      0      NaN  MnPrv      NaN
2  NaN      IR1      Lvl ...      0      NaN      NaN      Gar2
3  NaN      Reg      Lvl ...      0      NaN      NaN      NaN
4  NaN      IR1      Lvl ...      0      NaN  MnPrv      NaN

Misc Val  Mo Sold  Yr Sold  Sale Type  Sale Condition  SalePrice
0         0       5    2010      WD      Normal      215000
1         0       6    2010      WD      Normal      105000
2    12500       6    2010      WD      Normal      172000
3         0       4    2010      WD      Normal      244000
4         0       3    2010      WD      Normal      189900

[5 rows x 82 columns]
```

Entity: House

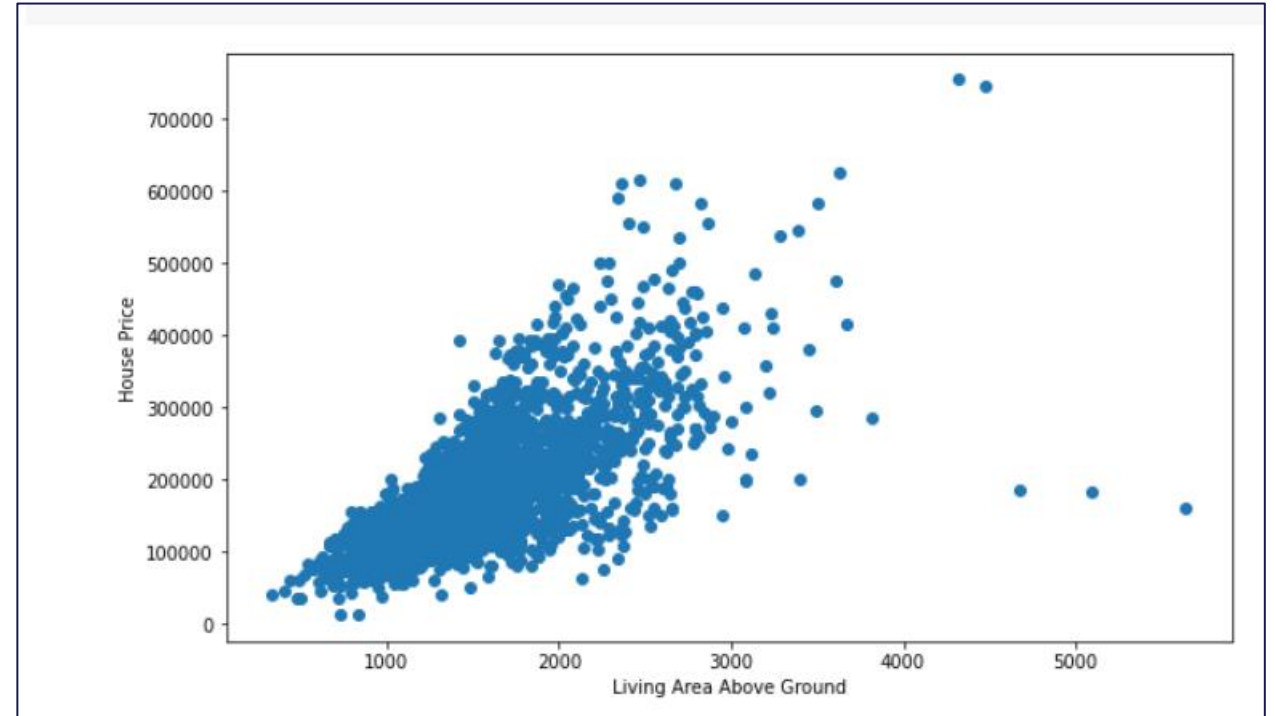
2 Characteristics: Living area + Price

```
[ ] #Page 217
```

```
fig, ax = plt.subplots(figsize=(10,6))

ax.scatter(x = df['Gr Liv Area'], y=df['SalePrice'])
plt.xlabel('Living Area Above Ground')
plt.ylabel('House Price')

plt.show()
```



2 Scatter Plots: Stacked

Scatter Plot1: Living Area / Price

Scatter Plot2: Overall Quality / Price



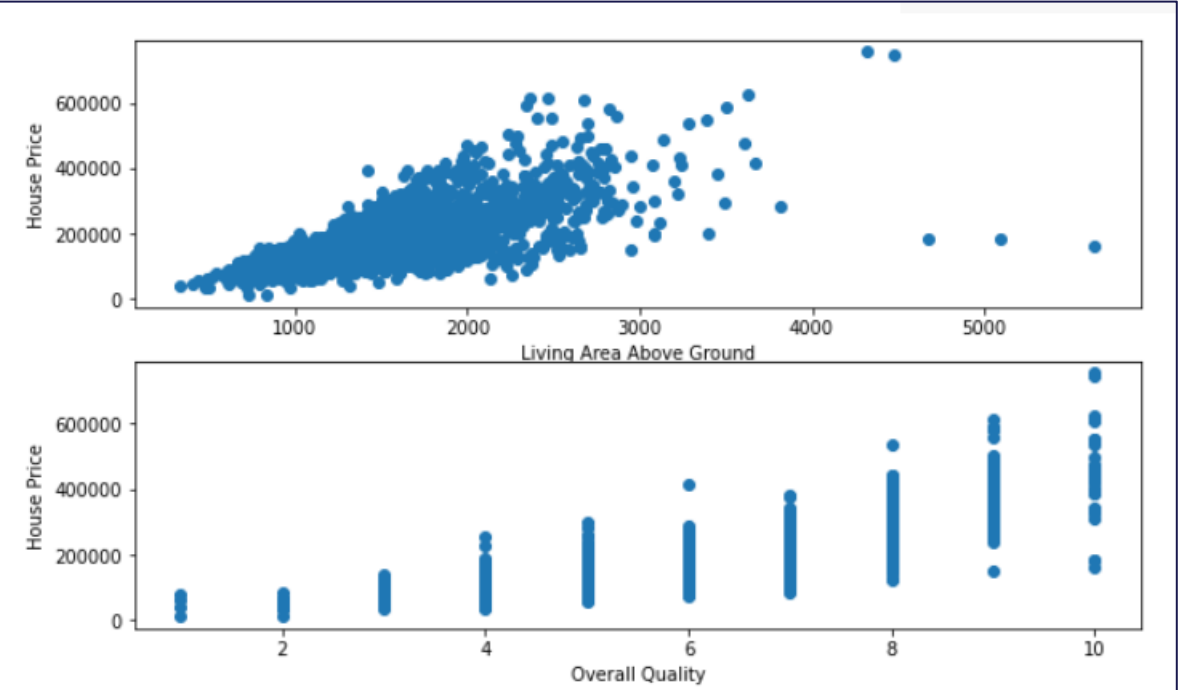
#Page 219

```
fig, ax = plt.subplots(2, figsize=(10,6))

ax[0].scatter(x = df['Gr Liv Area'], y=df['SalePrice'])
ax[0].set_xlabel('Living Area Above Ground')
ax[0].set_ylabel('House Price')

ax[1].scatter(x = df['Overall Qual'], y=df['SalePrice'])
ax[1].set_xlabel('Overall Quality')
ax[1].set_ylabel('House Price')

plt.show()
```



2 Scatter Plots: Stacked

Scatter Plot1: Living Area / Price

Scatter Plot2: Overall Quality / Price with a different marker

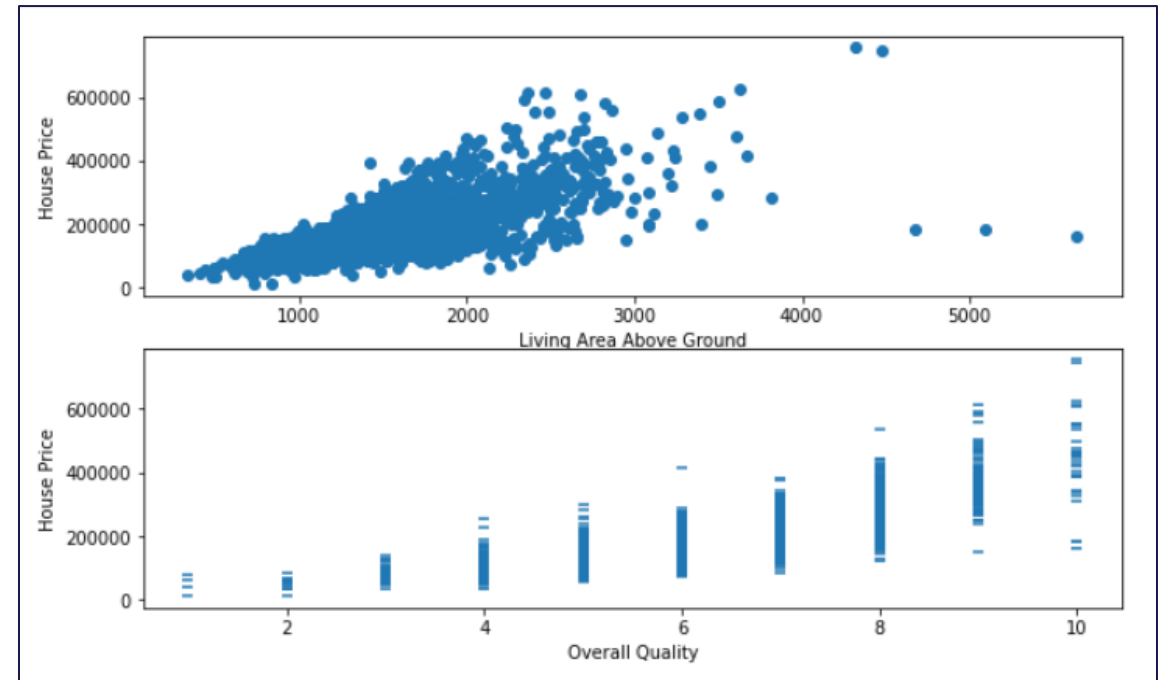
```
#Page 220

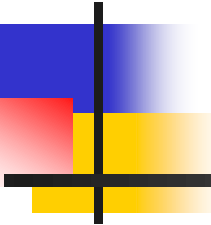
fig, ax = plt.subplots(2,figsize=(10,6))

ax[0].scatter(x = df['Gr Liv Area'], y=df['SalePrice'])
ax[0].set_xlabel('Living Area Above Ground')
ax[0].set_ylabel('House Price')

ax[1].scatter(x = df['Overall Qual'], y=df['SalePrice'], marker='_')
ax[1].set_xlabel('Overall Quality')
ax[1].set_ylabel('House Price')

plt.show()
```





Create 3D plot Using Scatter/Bubble Plot

Data Set: Homework#9

	A	B	C	D	E	F	G	
1		Country	Surface area (km2)	Population 1,000 (2017)	GDP (M US\$)	Life Exp Female	Life Exp Male	
2	1	Afghanistan	652864	35530	20270	63.5	61	
3	2	Albania	28748	2930	11541	79.9	75.6	
4	3	Algeria	2381741	41318	164779	76.5	74.1	
5	4	American Samoa	199	56	-99	77.8	71.1	
6	5	Angola	1246700	29784	117955	63	57.4	
7	6	Antigua and Barbuda	442	102	1356	78.2	73.3	
8	7	Argentina	2780400	44271	632343	79.8	72.2	
9	8	Armenia	29743	2930	10529	77	70.6	
10	9	Aruba	180	105	2702	77.8	72.9	
11	10	Australia	7692060	24451	1230859	84.4	80.2	
12	11	Austria	83871	8736	376967	83.5	78.4	
13	12	Azerbaijan	86600	9828	53049	74.6	68.6	
14	13	Bahamas	13940	395	8854	78.1	72	



Load Libraries

Upload Data File

Read Data File Using Pandas

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
%matplotlib inline
```

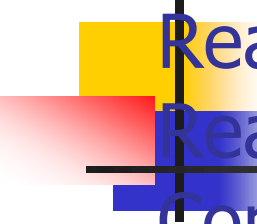
```
from google.colab import files
f = files.upload()
```

 No file chosen Upload widget is only available when the cell is in edit mode. Saving country_profile_variables.csv to country_profile_variables.csv

```
[ ] # Step# 1
#

country = pd.read_csv('country_profile_variables.csv')
print(country.head(5))
print(country.tail(5))
print(country.shape)

print(type(country))
```



Read Life Expectancy: Female

Read Life Expectancy: Male

Compute Difference between M/F
LE

Compute Average Life Expectancy

```
▶ lifeExpFemale = country['Life Exp Female']  
   lifeExpMale = country['Life Exp Male']  
  
   diffLE_MaleFemale = lifeExpFemale - lifeExpMale  
   averageLifeExp = (lifeExpFemale + lifeExpMale)/2  
   print(averageLifeExp)
```

```
① 0      62.25  
   1      77.75  
   2      75.30  
   3      74.45  
   4      60.20  
   ...  
  206     75.75  
  207     68.60  
  208     64.20  
  209     59.70  
  210     57.55  
   Length: 211, dtype: float64
```

Add Average Life Expectancy + Difference in the Country DataFrame

```
country['averageLifeExp'] = averageLifeExp
country['Difference'] = diffLE_MaleFemale

print(country.head(5))
```

	Unnamed: 0	Country	Surface area (km2)	Population 1,000	(2017)	\
0	1	Afghanistan	652864		35530	
1	2	Albania	28748		2930	
2	3	Algeria	2381741		41318	
3	4	American Samoa	199		56	
4	5	Angola	1246700		29784	

	GDP (M US\$)	Life Exp Female	Life Exp Male	averageLifeExp	Difference
0	20270	63.5	61.0	62.25	2.5
1	11541	79.9	75.6	77.75	4.3
2	164779	76.5	74.1	75.30	2.4
3	-99	77.8	71.1	74.45	6.7
4	117955	63.0	57.4	60.20	5.6

Sort 'country' Data Frame by Difference of Male/Female Life Expectancy

Country where the Male/Female life expectancy is highest is Syria

```
sortedTable = country.sort_values(['Difference'],ascending=False)
print(sortedTable.head(5))
print(sortedTable.shape)
sortedTable.loc[sortedTable['Country'] == 'United States of America']
```

```
↳ Unnamed: 0      Country  Surface area (km2) \
182      183  Syrian Arab Republic      185180
154      155   Russian Federation     17098246
16        17         Belarus      207600
107      108         Lithuania      65286
102      103          Latvia      64573

      Population 1,000 (2017)  GDP (M US$)  Life Exp Female  Life Exp Male \
182                    18270      28393          76.3          64.4
154                   143990     1326016          75.9          64.7
16                     9468      54609          77.7          66.5
107                    2890      41402          79.3          68.5
102                    1950      27004          78.7          68.8

      averageLifeExp  Difference
182                70.35        11.9
154                70.30        11.2
16                 72.10        11.2
107                73.90        10.8
102                73.75         9.9
(211, 9)
```

Create 3D plot + Color

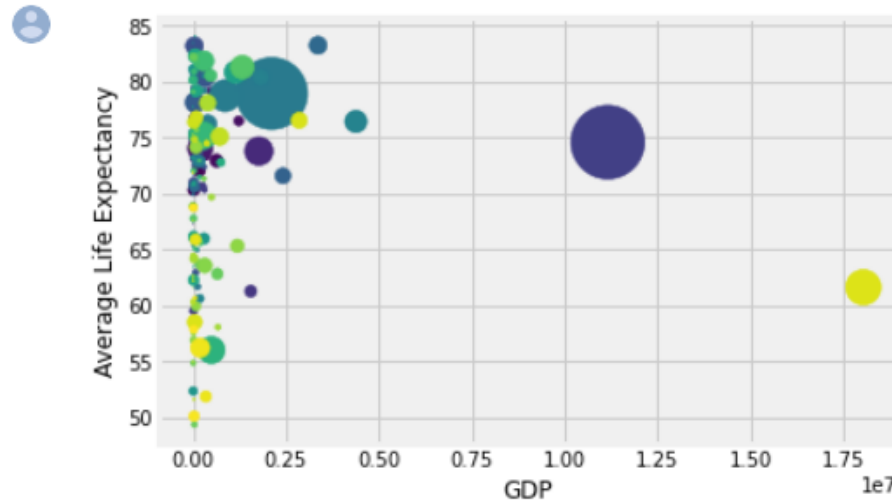
X-axis: GDP

Y-axis: Avg. Life Expectancy

**Size of the bubble is
Proportional to the
Population**

```
countryName = country['Country']  
GDP = country['GDP (M US$)']  
averageLifeExp = sortedTable['averageLifeExp']  
  
marker_size_P = country['Population 1,000 (2017)']/1000 # Proportional to the Population  
marker_size_SA = country['Surface area (km2)']/10000 # Proportional to the Surface Area
```

```
fig, ax = plt.subplots()  
  
ax.scatter(GDP, averageLifeExp, s=marker_size_P, c = range(len(countryName)))  
ax.set_xlabel("GDP")  
ax.set_ylabel("Average Life Expectancy")  
plt.show()
```



Create 3D plot + Color

X-axis: GDP

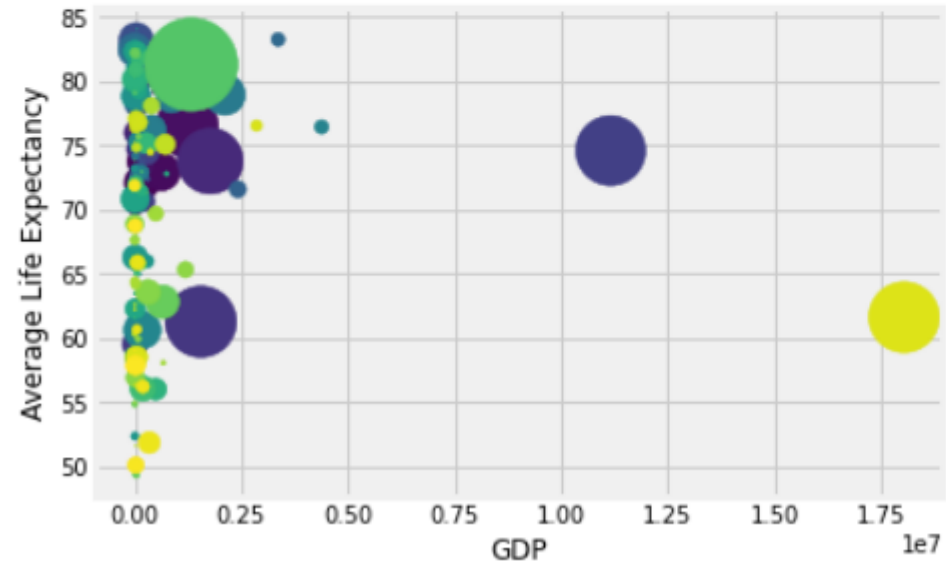
Y-axis: Avg. Life Expectancy

**Size of the bubble is
Proportional to the Surface
Area of the Country**

```
fig, ax = plt.subplots()

ax.scatter(GDP, averageLifeExp, s=marker_size_SA, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```

 /usr/local/lib/python3.9/dist-packages/matplotlib/collections.py:963: RuntimeWarning
scale = np.sqrt(self._sizes) * dpi / 72.0 * self._factor



Create 3D plot + Color

X-axis: $\log(\text{GDP})$

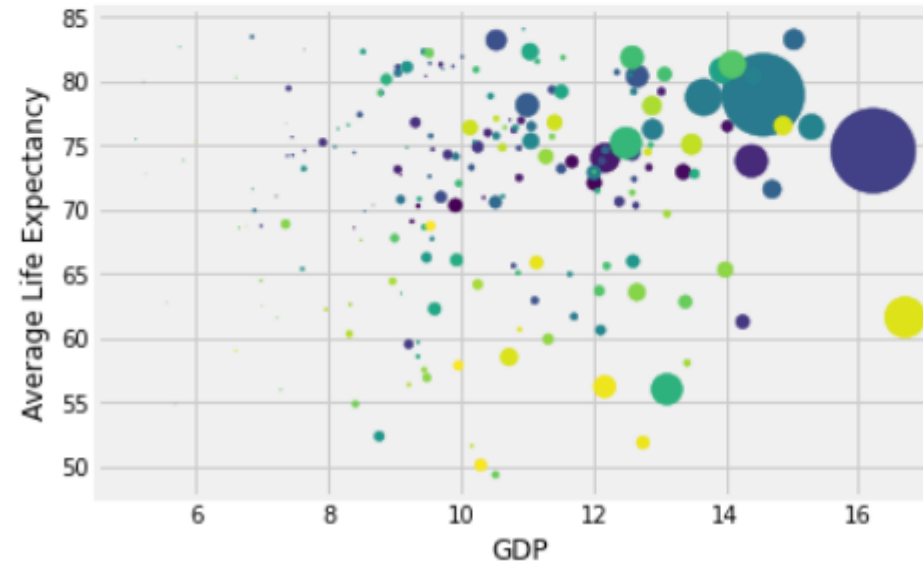
Y-axis: Avg. Life Expectancy

**Size of the bubble is
Proportional to the
Population**

```
fig, ax = plt.subplots()

ax.scatter(np.log(GDP), averageLifeExp, s=marker_size_P, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```

 /usr/local/lib/python3.9/dist-packages/pandas/core/arraylike.py:397: RuntimeWarning: result = getattr(ufunc, method)(*inputs, **kwargs)



Create 3D plot + Color

X-axis: $\log(\text{GDP})$

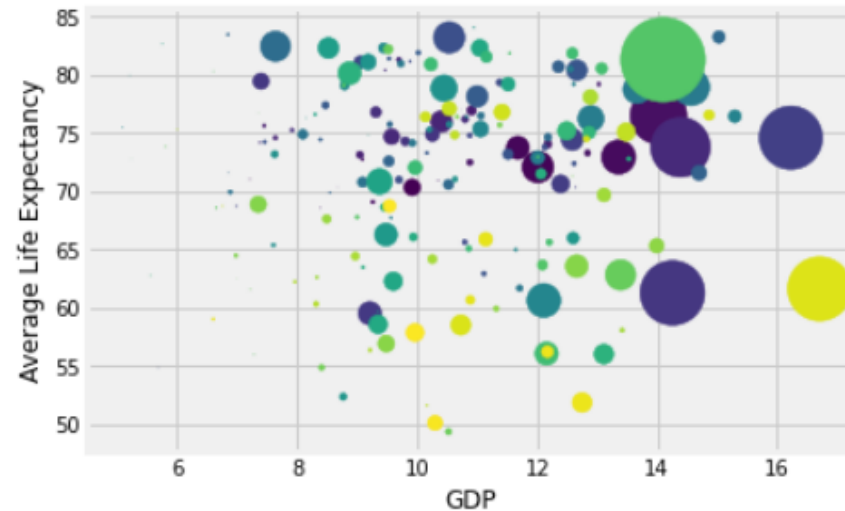
Y-axis: Avg. Life Expectancy

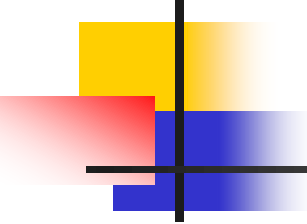
**Size of the bubble is
Proportional to the Surface
Area of the Country**

```
fig, ax = plt.subplots()

ax.scatter(np.log(GDP), averageLifeExp, s=marker_size_SA, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```

```
/usr/local/lib/python3.9/dist-packages/pandas/core/arraylike.py:397: RuntimeWarning: in
result = getattr(ufunc, method)(*inputs, **kwargs)
/usr/local/lib/python3.9/dist-packages/matplotlib/collections.py:963: RuntimeWarning: in
scale = np.sqrt(self._sizes) * dpi / 72.0 * self._factor
```



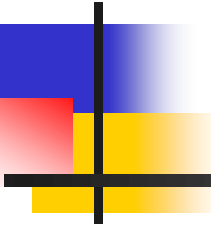


Outline

- When do we need to use Scatter Plots?
- Data Set: Ames Iowa Housing
- Create Scatter Plot: Living Area + Price of House
- Create 3D plot Using Scatter/Bubble Plot + Color
 - GDP & Life Expectance of Countries

Statistical Computing and Data Visualization

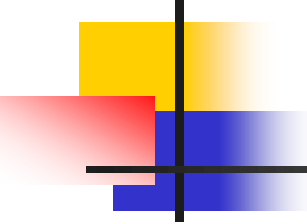
DSO 545



Dr. Ash Pahwa

Lesson 8: Bar + Pie + Scatter Plots

Lesson 8.5: Histogram Plots



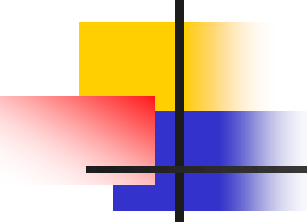
Plot Characteristics

- Change the Figure Size
- Adding Colors + Color Map
- Adding labels on plots
- Add Legend
- Font Size
- Set Axis Range
- Change Tick Frequency
- Rotate Axis Tick Labels
- Draw Vertical and Horizontal Lines on Plot



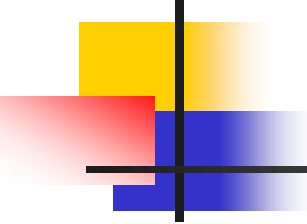
Type of Plots

1. Line Plot
2. Bar Plot
3. Pie Chart
4. Scatter Plot
5. Histogram Plot
6. Box Plot
7. Joint Plot
8. Violin Plots
9. Stack Plots
10. Heat Maps
11. Spectrogram Plots
12. 3D Plots
13. Contour Plots
14. KDE Plots
15. Interactive Plots



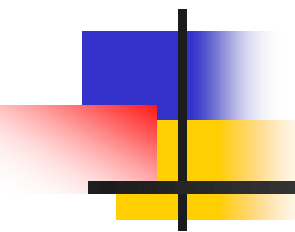
Outline

- Plot Histogram – Netflix Data
- Changing the Bin Size
- Fixed number of Bins
- KDE Plots
- Overlay KDE with Histogram
- Histogram of some Raw data in a list



Histograms

- Histogram displays the Frequency of data
- Used to visualize distribution of data
- Data is placed into ranges: called bins or buckets
- Histogram plots can be overlaid with KDE (Kernel Density Estimation) line



Data File

Netflix_titles.csv

	A	B	C	D	E	F	G	H	I	J	K	L
1	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descripti
2	s1	Movie	Dick Johns	Kirsten Johnson		United Sta	25-Sep-21	2020	PG-13	90 min	Document	As her fa
3	s2	TV Show	Blood & Water		Ama Qama	South Afric	24-Sep-21	2021	TV-MA	2 Seasons	Internatio	After cro
4	s3	TV Show	Ganglands	Julien Lecl	Sami Bouajila, Tracy C		24-Sep-21	2021	TV-MA	1 Season	Crime TV S	To prote
5	s4	TV Show	Jailbirds New Orleans				24-Sep-21	2021	TV-MA	1 Season	Docuseries	Feuds, fli
6	s5	TV Show	Kota Factory		Mayur Mo	India	24-Sep-21	2021	TV-MA	2 Seasons	Internatio	In a city
7	s6	TV Show	Midnight M	Mike Flana	Kate Siegel, Zach Gilfo		24-Sep-21	2021	TV-MA	1 Season	TV Dramas	The arriv
8	s7	Movie	My Little P	Robert Cul	Vanessa Hudgens, Kin		24-Sep-21	2021	PG	91 min	Children & Equestria	
9	s8	Movie	Sankofa	Haile Gerir	Kofi Ghana	United Sta	24-Sep-21	1993	TV-MA	125 min	Dramas, In	On a pho
10	s9	TV Show	The Great	Andy Devc	Mel Giedro	United Kin	24-Sep-21	2021	TV-14	9 Seasons	British TV S	A talente
11	s10	Movie	The Starlin	Theodore	Melissa M	United Sta	24-Sep-21	2021	PG-13	104 min	Comedies, A woman	
12	s11	TV Show	Vendetta: Truth, Lies and The Mafia				24-Sep-21	2021	TV-MA	1 Season	Crime TV S	Sicily bo

Load Libraries + Upload and Read Data File

```
✓ [1] #1
0s import numpy as np
import matplotlib.pyplot as plt

✓ [2] from google.colab import files
26s f = files.upload()

Choose Files netflix_titles.csv
• netflix_titles.csv(text/csv) - 3399671 bytes, last modified: 3/26/2023 - 100% done
Saving netflix_titles.csv to netflix_titles.csv

✓ [3] import pandas as pd
0s df = pd.read_csv('netflix_titles.csv')

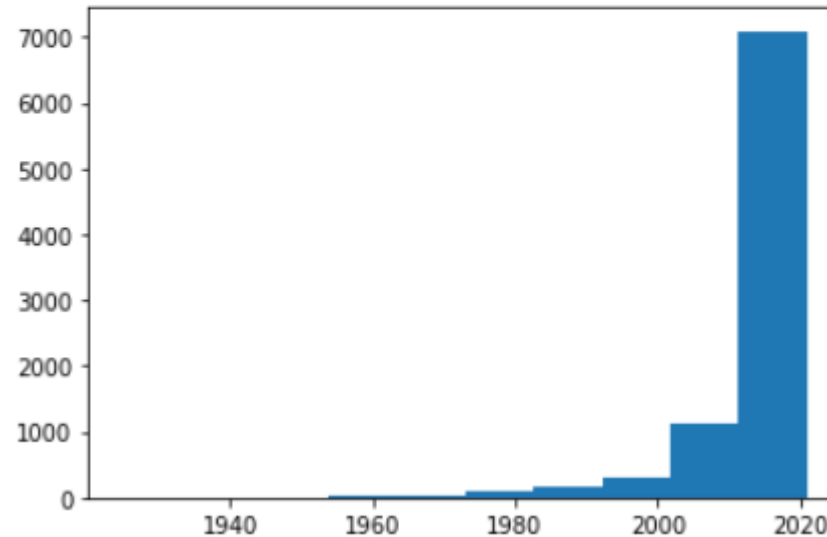
✓ [4] print(df.head())
0s
```

Simple Histogram



page 238

```
plt.hist(df['release_year'])  
plt.show()
```

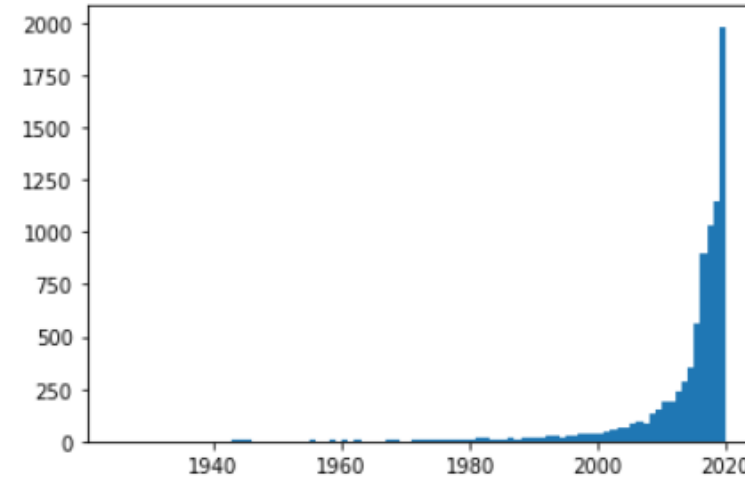


Change Bin Size

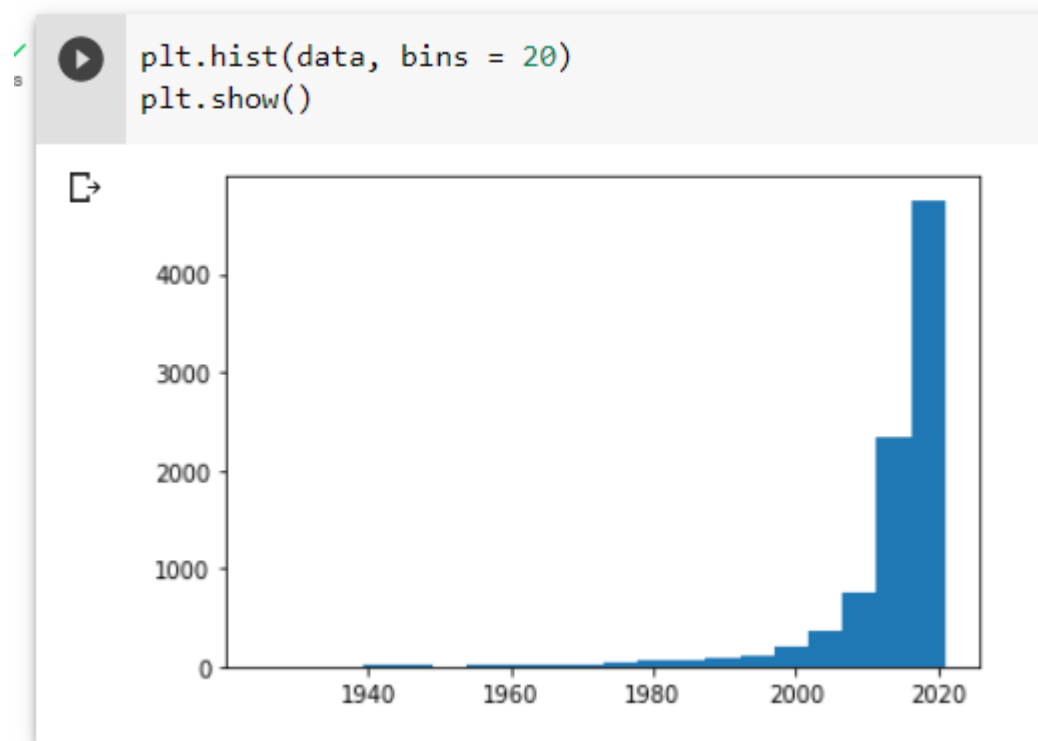
```
# page 240
data = df['release_year']
print(np.arange(min(data), max(data)))

plt.hist(data, bins = np.arange(min(data), max(data), 1))
plt.show()
```

```
[1925 1926 1927 1928 1929 1930 1931 1932 1933 1934 1935 1936 1937 1938
1939 1940 1941 1942 1943 1944 1945 1946 1947 1948 1949 1950 1951 1952
1953 1954 1955 1956 1957 1958 1959 1960 1961 1962 1963 1964 1965 1966
1967 1968 1969 1970 1971 1972 1973 1974 1975 1976 1977 1978 1979 1980
1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994
1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008
2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020]
```



Fixed Number of Bins



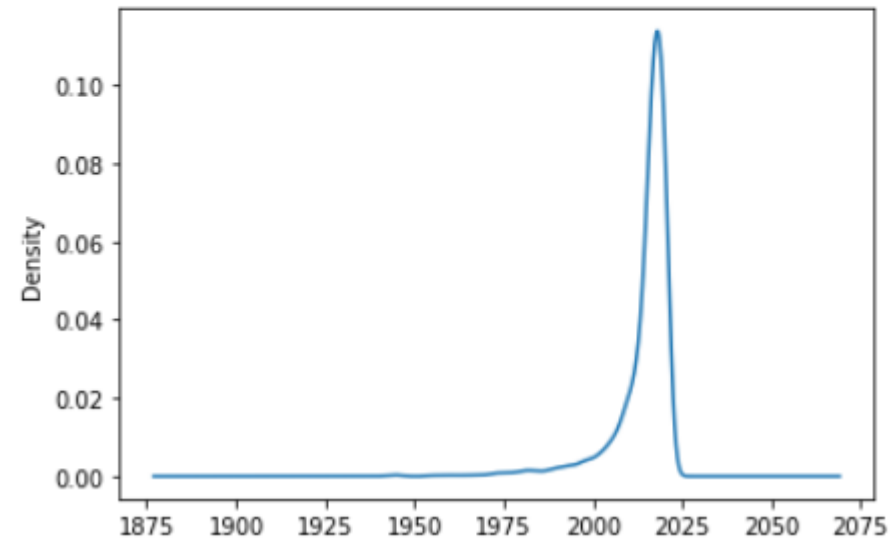
KDE Plot – Kernel Density Estimation



#page = 249

```
fig, ax = plt.subplots()
```

```
df['release_year'].plot.kde(ax=ax)  
plt.show()
```



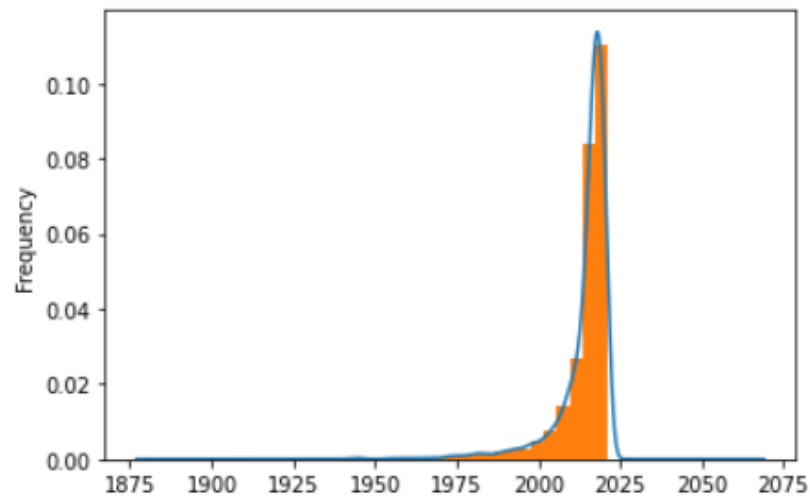
Overlay KDE with Histogram

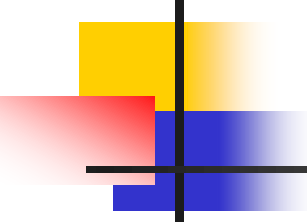


#page = 249

```
fig, ax = plt.subplots()
df['release_year'].plot.kde(ax=ax)

df['release_year'].plot.hist(ax=ax, bins=25, density=True)
plt.show()
```





Histogram of Raw Data in a List

```
[1] from matplotlib import pyplot as plt

[2] population_ages = [22,55,62,45,21,22,34,42,4,99,102,110,120,121,122,130,111,115,112,80,
                      75,65,54,44,43,42,48]

sorted_list = sorted(population_ages)
print(sorted_list)

[4, 21, 22, 22, 34, 42, 42, 43, 44, 45, 48, 54, 55, 62, 65, 75, 80, 99, 102, 110, 111, 112, 115, 120, 121, 122, 130]
```

Histogram

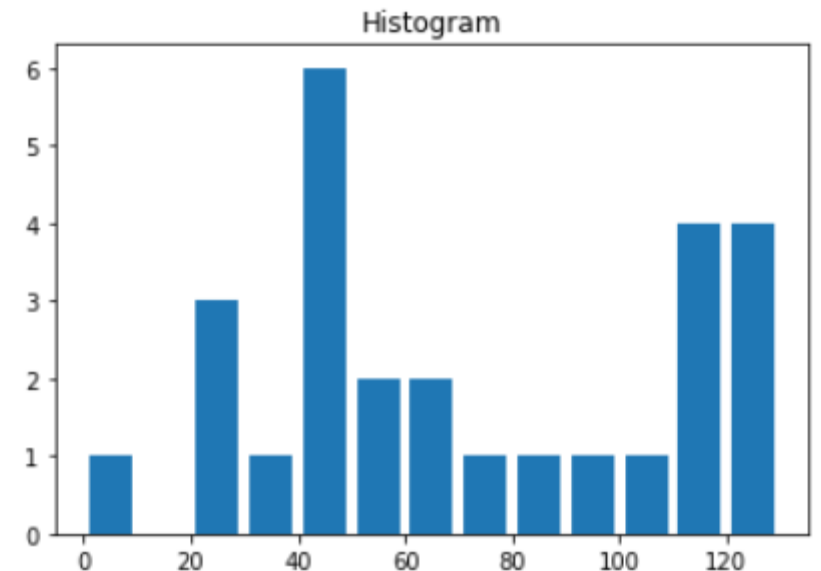
```
[2] population_ages = [22,55,62,45,21,22,34,42,4,99,102,110,120,121,122,130,111,115,112,80,  
                      75,65,54,44,43,42,48]
```

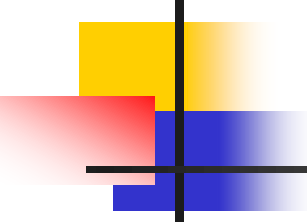
```
sorted_list = sorted(population_ages)  
print(sorted_list)
```

```
[4, 21, 22, 22, 34, 42, 42, 43, 44, 45, 48, 54, 55, 62, 65, 75, 80, 99, 102, 110, 111, 112, 115, 120, 121, 122, 130]
```

```
bins = [0,10,20,30,40,50,60,70,80,90,100,110,120,130]
```

```
plt.hist(population_ages, bins, histtype='bar',rwidth=0.8)  
plt.title('Histogram')  
plt.show()
```





Summary

- Plot Histogram – Netflix Data
- Changing the Bin Size
- Fixed number of Bins
- KDE Plots
- Overlay KDE with Histogram
- Histogram of some Raw data in a list